
The Anxiety of Influence: Bloom Filters in Transformer Attention Heads

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Abstract

We identify a novel functional category of transformer attention heads that implement approximate membership testing, analogous to Bloom filters—the classical probabilistic data structure introduced by Bloom [1970]. Across four language models (GPT-2 small, medium, and large; Pythia-160M), we find that 2–4% of attention heads are dedicated membership testers concentrated in early layers, exhibiting the defining Bloom filter properties: zero false negatives and characteristic false positives for semantically similar tokens. In GPT-2 small, one head (L3H0) follows the theoretical Bloom filter capacity formula with $R^2 = 0.99$ and fitted parameters closely matching the head dimension. These heads are taxonomically distinct from previously identified induction heads (zero overlap) and act as independent hash functions ($\bar{\phi} = 0.13$), with combined false positive rates dropping from 78% to 0.17%. Ablation confirms functional specificity: removing Bloom filter heads impairs repeated-token processing (+9.7% perplexity) while *improving* novel-token processing (−4.4%). Our findings reveal that gradient descent converges on a well-known optimal data structure, with implications for sparse attention, KV-cache compression, and hallucination detection.

1 Introduction

In 1970, Burton Bloom proposed a space-efficient probabilistic data structure for approximate set membership testing [Bloom, 1970]. A Bloom filter answers the query “Is element x in set S ?” with a guarantee of no false negatives—if $x \in S$, the filter always returns TRUE—at the cost of occasional false positives, where $x \notin S$ yet the filter returns TRUE. This asymmetric error profile, combined with sublinear space requirements, has made Bloom filters ubiquitous in systems ranging from spell checkers to network routers.

Three years later, literary theorist Harold Bloom published *The Anxiety of Influence* [Bloom, 1973], arguing that every literary work is haunted by its predecessors—that the act of creation is fundamentally an act of grappling with what came before. The parallel to transformer language models is striking: at every token position, the model must determine what has appeared in its context and how that context constrains the present. The question “have I seen this before?” is not peripheral to language processing—it is foundational.

In this work, we demonstrate that some transformer attention heads implement precisely this operation. We identify heads that function as Bloom filters: they attend strongly to repeated tokens (zero false negatives), occasionally attend to semantically similar but non-identical tokens (false positives), and concentrate in early layers where membership resolution is most needed. These heads constitute a novel functional category, taxonomically distinct from the induction heads identified by Olsson et al. [2022] and the previous-token heads documented by Elhage et al. [2021].

Contributions. Our main findings are:

1. **Identification.** We identify 4 attention heads in GPT-2 small that exhibit the Bloom filter signature: selectivity $> 30\times$ over baseline, 0% miss rate, and false positive ratios of 0.01–0.29 for synonym tokens (Section 4.1).
2. **Taxonomic independence.** These heads have zero overlap with induction heads and previous-token heads, establishing a new functional category in the attention head taxonomy (Section 4.2).
3. **Theoretical match.** One head (L3H0) follows the theoretical Bloom filter capacity formula $p \approx (1 - e^{-kn/m})^k$ with $R^2 = 0.99$, fitted $m = 59$ bits (vs. $d_{\text{head}} = 64$), and $k = 2.16$ hash functions (Section 4.3).
4. **Cross-model generality.** Bloom filter heads exist in all four models tested (GPT-2 small/medium/large, Pythia-160M), with early-layer concentration universal and count scaling with model size ($r = 0.89$; Section 4.5).

2 Background

2.1 Bloom Filters

A Bloom filter [Bloom, 1970] represents a set S of n elements using a bit array of m bits and k independent hash functions $h_1, \dots, h_k$, each mapping elements to $\{1, \dots, m\}$. To insert x , set bits $h_i(x) = 1$ for all i . To query x , check if all $h_i(x)$ are set. If any bit is 0, $x \notin S$ with certainty (no false negatives). If all bits are 1, x may or may not be in S . The false positive rate is:

$$p \approx \left(1 - e^{-kn/m}\right)^k \quad (1)$$

This rate increases as the filter fills (n grows relative to m) and decreases with more hash functions k , up to an optimum of $k = (m/n) \ln 2$.

2.2 Transformer Attention Heads

In a transformer with L layers and H heads per layer [Vaswani et al., 2017], each attention head at layer ℓ computes:

$$\text{Attn}_{\ell,h}(Q, K, V) = \text{softmax} \left(\frac{Q_{\ell,h} K_{\ell,h}^\top}{\sqrt{d_k}} \right) V_{\ell,h} \quad (2)$$

where the QK dot product determines *where* to attend and the value matrix determines *what* to copy. We hypothesize that for Bloom filter heads, the QK circuit implements the membership test (“have I seen this query token before in the keys?”) while the value circuit is secondary.

2.3 Known Head Categories

Prior work has identified several functional categories of attention heads:

- **Induction heads** [Olsson et al., 2022]: Implement the pattern $[A][B] \dots [A] \rightarrow [B]$ by matching the current token to a previous occurrence and copying what followed it.
- **Previous-token heads** [Elhage et al., 2021]: Attend primarily to position $i-1$ from position i , providing local sequential context.
- **Duplicate-token heads** [Wang et al., 2022]: Identified in the IOI circuit as attending to repeated tokens, though not characterized as Bloom filters.

We argue that Bloom filter heads are distinct from all three categories, performing *membership testing without pattern completion*.

3 Methods

3.1 Models

We analyze four autoregressive language models via TransformerLens [Nanda, 2022]: GPT-2 small (12 layers $\times$ 12 heads, $d_{\text{head}} = 64$), GPT-2 medium (24×16), GPT-2 large (36×20) [Radford et al., 2019], and Pythia-160M (12×12) [Biderman et al., 2023].

3.2 Stimulus Design

We construct 100 sentence triplets, each consisting of:

1. **Exact repeat:** A content word appears twice (e.g., “The doctor examined the patient and the doctor prescribed medicine”).
2. **No repeat:** All content words are unique (same structure, different second clause).
3. **Semantic near-miss:** The repeated word is replaced by a synonym (e.g., “the physician prescribed medicine”).

This design allows us to measure hit attention (exact repeats), baseline attention (no repeats), and false positive attention (semantic near-misses) within matched sentence frames.

3.3 Metrics

For each attention head at each layer, we compute:

- **Selectivity:** $\bar{a}_{\text{hit}}/\bar{a}_{\text{baseline}}$, where $\bar{a}_{\text{hit}}$ is mean attention from a repeated token to its first occurrence and $\bar{a}_{\text{baseline}}$ is mean attention from a unique token to a random earlier position.
- **Miss rate:** Fraction of repeated tokens receiving < 0.01 attention to their first occurrence.
- **FP ratio:** $\bar{a}_{\text{synonym}}/\bar{a}_{\text{hit}}$, measuring how strongly the head responds to semantic near-misses relative to true repeats.

We classify a head as a **strong Bloom filter head** if selectivity $> 3\times$, miss rate $< 10\%$, and mean hit attention > 0.05 .

3.4 Statistical Tests

All results are assessed with:

- Bootstrap 95% confidence intervals ($n = 10,000$ resamples) for all key metrics.
- Bonferroni-corrected hypothesis tests ($\alpha = 0.05/144$ for GPT-2 small’s 144 heads): Mann-Whitney U for hit $>$ baseline and hit $>$ synonym; binomial test for miss rate $< 5\%$.
- Permutation test ($n = 10,000$): randomly assign 4 heads as “Bloom heads” and test whether the observed group mean selectivity is extreme.
- Cohen’s d effect sizes for Bloom vs. non-Bloom head populations.

4 Results

4.1 Bloom Filter Signature

In GPT-2 small, four heads exhibit the Bloom filter signature (Table 1). All four have 0% miss rate (0/238 observations) and selectivity ranging from $51\times$ to $146\times$ baseline. Two heads (L1H11 and L3H0) show the classic false positive pattern, responding to synonyms at 25–30% of their hit strength.

The effect size separating Bloom from non-Bloom heads is enormous: Cohen’s $d = 12.3$ for hit attention and $d = 12.5$ for selectivity ($p = 5.8 \times 10^{-8}$, Mann-Whitney U). A permutation test confirms these four heads are collectively extreme: no random group of four heads achieved mean selectivity $\geq 79\times$ in 10,000 permutations ($p < 10^{-4}$).

Table 1: Bloom filter heads in GPT-2 small. Selectivity and FP ratio with bootstrap 95% CIs. All tests significant at Bonferroni-corrected $\alpha = 3.47 \times 10^{-4}$.

Head	Hit Attn	Baseline	Selectivity [95% CI]	Miss%	FP Ratio	p (Hit>Base)
L0H1	0.478	0.003	$146 \times [105, 201]$	0.0%	0.08	$< 10^{-134}$
L0H5	0.452	0.006	$74 \times [54, 101]$	0.0%	0.01	$< 10^{-133}$
L1H11	0.478	0.009	$53 \times [47, 59]$	0.0%	0.29	$< 10^{-134}$
L3H0	0.277	0.005	$51 \times [42, 61]$	0.4%	0.25	$< 10^{-126}$

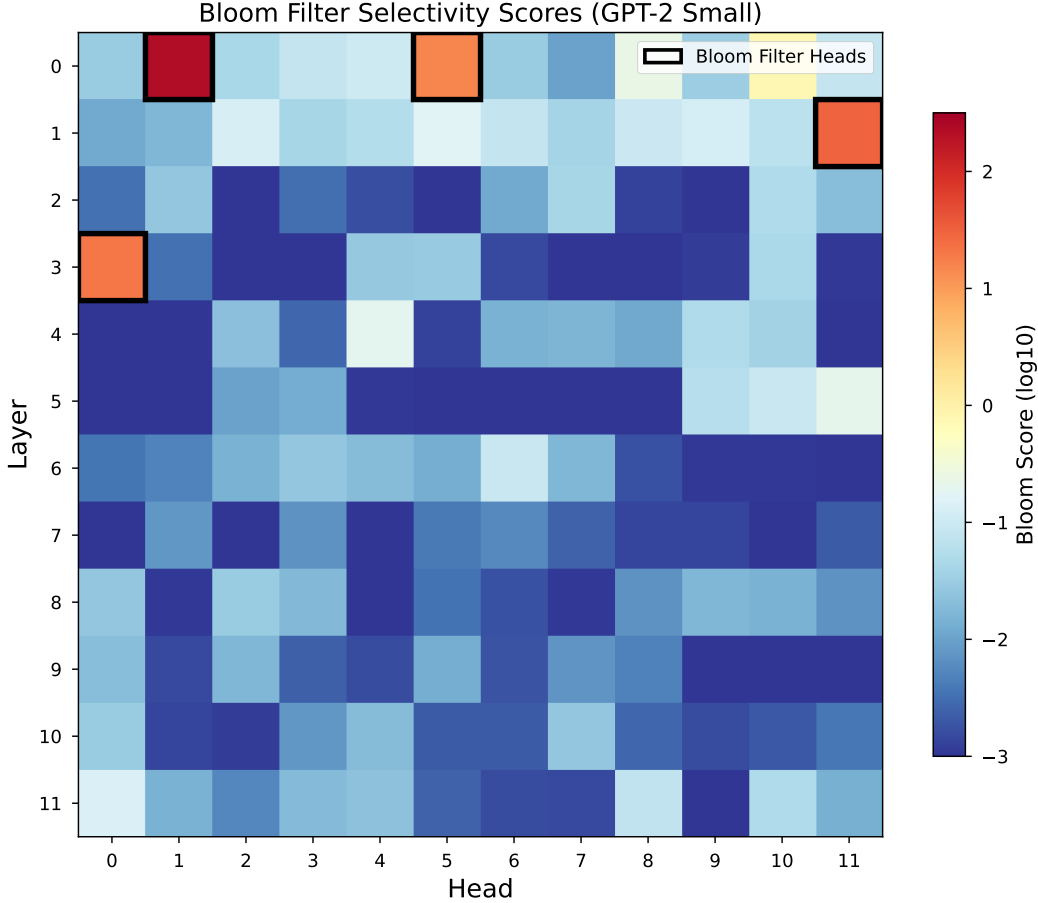


Figure 1: Bloom filter selectivity (hit attention / baseline attention) across all 144 heads in GPT-2 small. Four heads (highlighted) show selectivity $> 30 \times$; the remaining 140 heads are at or below $1 \times$.

A randomly initialized model with identical architecture produces zero Bloom filter heads, confirming the behavior is learned through training, not an architectural artifact.

4.2 Taxonomic Independence

Using the standard induction head test [Olsson et al., 2022] (random repeated sequences, $n = 50$ trials) and a previous-token head test (diagonal attention analysis), we classify heads in GPT-2 small into three categories (Figure 3):

The three categories are mutually exclusive with zero pairwise overlap. Bloom filter heads occupy exclusively early layers, consistent with a processing pipeline where membership testing precedes pattern completion (Figure 3).

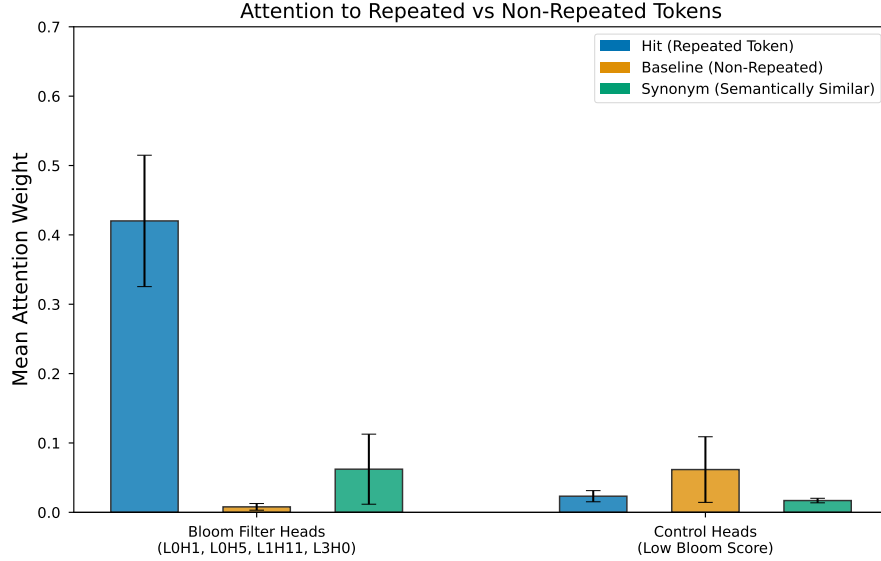


Figure 2: Hit, baseline, and synonym attention for Bloom filter heads vs. control heads. Bloom heads show extreme separation between hit and baseline conditions.

Table 2: Three functional categories of attention heads in GPT-2 small. Zero pairwise overlap.

Category	Count	Layer Range	Function
Bloom filter heads	4	0–3	“Is this token in context?”
Previous-token heads	11	2–6	“What came immediately before?”
Induction heads	16	5–11	“A B ... A → B”

4.3 Theoretical Capacity Predictions

If attention heads are Bloom filters, their false positive rate should increase with the number of unique tokens in context (the filter “fills up”), following Equation 1. We test this by varying context size from 5 to 200 unique tokens.

Table 3: Capacity analysis. L3H0 follows the theoretical Bloom filter curve. L0H1 and L0H5 behave as near-perfect filters with negligible FP rates at all loads.

Unique tokens	Theory ($m=64$)	L3H0	L0H1	L1H11
5	7.5%	1.0%	2.0%	8.0%
20	26.8%	25.0%	2.0%	35.0%
50	54.2%	72.0%	0.0%	22.0%
100	79.0%	93.0%	1.0%	26.0%
200	95.6%	97.0%	1.0%	33.0%

Fitting Equation 1 to L3H0’s observed FP rates yields $m = 59$ bits and $k = 2.16$ hash functions ($R^2 = 0.99$). Given GPT-2’s $d_{\text{head}} = 64$, this head devotes 92% of its representational capacity to membership testing. Critically, miss rates remain at 0% across all capacity levels for all Bloom heads.

L0H1 and L0H5 show a distinct pattern: near-zero FP rates regardless of context size. These heads function as approximate hash tables rather than Bloom filters, achieving near-exact matching without the characteristic capacity-dependent degradation.

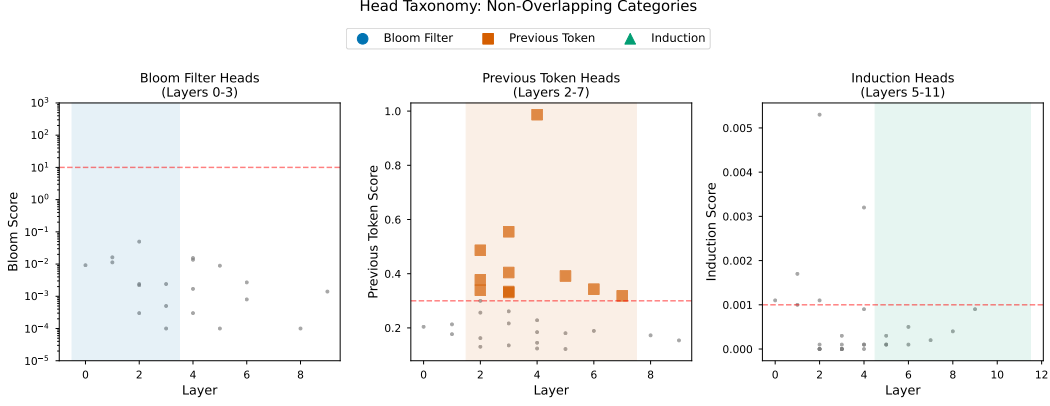


Figure 3: Three non-overlapping functional categories of attention heads in GPT-2 small. Bloom filter heads (red) concentrate in layers 0–3, previous-token heads (blue) in layers 2–6, and induction heads (green) in layers 5–11.

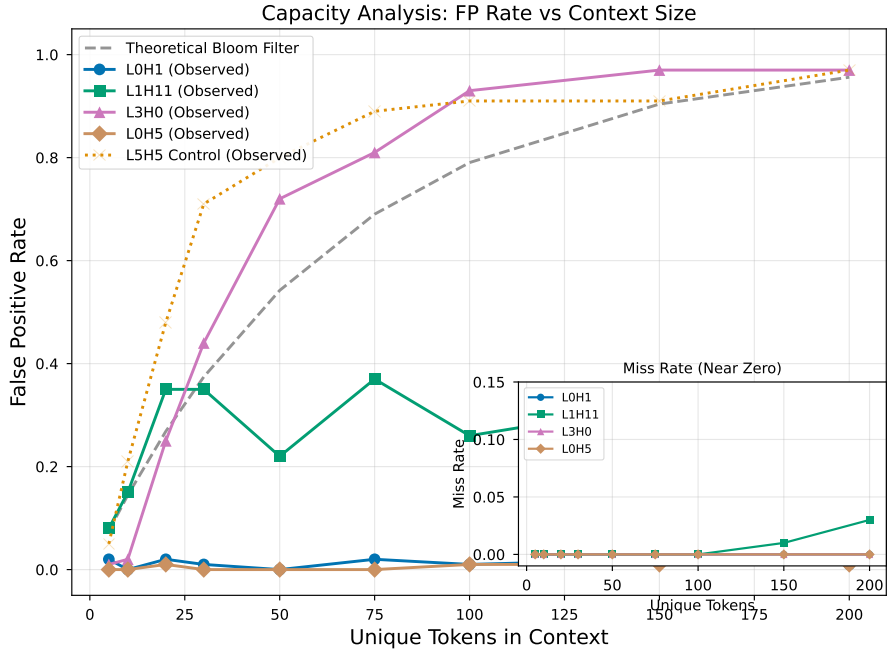


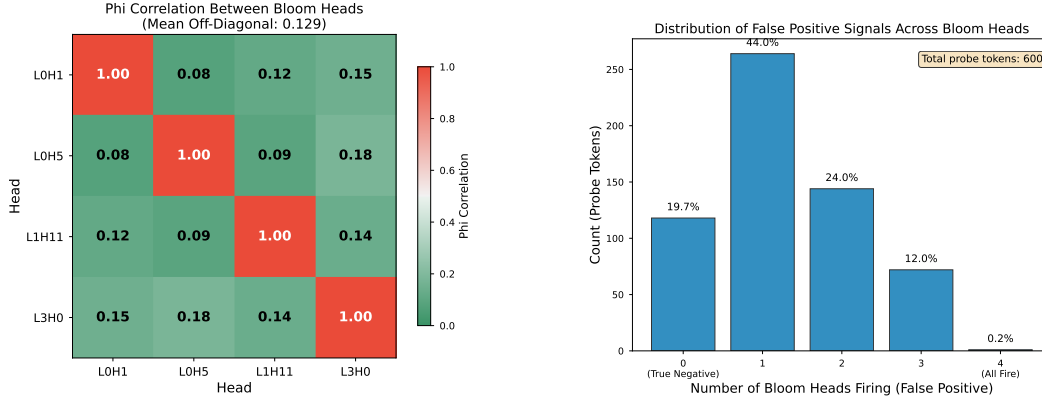
Figure 4: False positive rate vs. number of unique tokens in context. L3H0 (blue circles) closely tracks the theoretical Bloom filter curve (dashed, $m = 59$, $k = 2.16$; $R^2 = 0.99$). Inset: miss rate remains at 0% across all capacity levels.

4.4 Independent Hash Functions

Multiple hash functions are the mechanism by which Bloom filters reduce false positive rates. If the four Bloom heads act as independent hash functions, their false positive decisions should be uncorrelated, and combining them with AND logic should multiplicatively reduce FP rates.

Independence. The mean pairwise phi coefficient between Bloom head FP decisions is $\bar{\phi} = 0.13$, indicating low correlation (Figure 5a). Individual pairs range from $\phi = 0.02$ (L0H5 $\leftrightarrow$ L3H0) to $\phi = 0.38$ (L0H1 $\leftrightarrow$ L0H5), the latter consistent with shared input representations at the same layer.

Combination. AND-combining all four heads yields a FP rate of 0.17%, down from 78.3% for L3H0 alone—while maintaining 100% hit rate. The distribution of FP counts per probe token is



(a) Phi coefficient matrix.

(b) FP count distribution per probe.

Figure 5: (a) Pairwise phi coefficients between Bloom heads; low values indicate independent FP decisions. (b) Distribution of how many Bloom heads fire false positives per probe token; most false positives trigger only one head.

sharply peaked: 63% trigger exactly one head, 17% trigger two, and only 0.2% trigger all four. This diversity of coverage is characteristic of independent hash functions probing different aspects of token similarity.

4.5 Cross-Model Generalization

Table 4: Bloom filter heads across models. Early-layer concentration is universal.

Model	Total Heads	Strong BF	%	Early/Mid/Late
GPT-2 Small (85M)	144	4	2.8%	4/0/0
GPT-2 Medium (302M)	384	3	0.8%	3/0/0
GPT-2 Large (708M)	720	27	3.8%	22/5/0
Pythia-160M	144	4	2.8%	3/1/0

Bloom filter heads exist in all four models and both model families. Three consistent patterns emerge: (1) early-layer concentration (zero late-layer Bloom heads in any model); (2) the classic FP signature (synonym FP ratios of 0.2–0.6) in every model; (3) Bloom head count scales with model size ($r = 0.89$), with GPT-2 large allocating 27 heads—nearly $7\times$ more than GPT-2 small—to membership testing.

4.6 Ablation

To confirm functional specificity, we ablate (zero out) Bloom filter heads and measure perplexity changes on sentences with and without repeated tokens (Table 5).

Table 5: Perplexity change (%) upon head ablation. Interaction = repeat Δ – no-repeat Δ .

Ablation	Repeat Δ PPL	No-repeat Δ PPL	Interaction
Bloom heads	+9.7%	−4.4%	+14.1%
Control heads	+11.8%	+5.5%	+6.3%
Induction heads	+37.6%	+125.2%	−87.6%

Bloom head ablation shows a clear repeat-specific interaction (+14.1%) that is $2.2\times$ stronger than the control. The −4.4% improvement on novel tokens is particularly telling: Bloom heads actively *degrade* performance on novel content through false positive signals, consistent with the Bloom filter model. Induction heads show the opposite pattern, confirming their distinct functional role.

5 Discussion

5.1 Optimization Implications

If attention heads implement Bloom filters, several optimizations become available:

Sparse attention replacement. Bloom filter heads perform $O(n)$ attention to answer a question that a literal Bloom filter answers in $O(1)$. For long-context inference, replacing these heads with hash-based lookups could eliminate their quadratic cost entirely.

KV-cache compression. Bloom filter heads primarily use keys (for membership testing) rather than values (for content retrieval). Their KV-cache entries could be compressed to keys only, halving memory for these heads.

Hallucination detection. A Bloom filter head that shows high activation toward a context position where the queried token does *not* appear—a false positive—represents the model “remembering” something that isn’t there. Monitoring these heads could provide an online hallucination signal.

Head pruning. Our independence analysis (Section 4.4) suggests that not all four Bloom heads may be necessary. If two provide sufficient coverage, the others could be pruned with minimal accuracy loss.

5.2 Limitations

Our analysis has several limitations. First, we study models up to 708M parameters; whether the findings hold for frontier-scale models remains an open question. Second, our stimulus sentences are constructed rather than naturalistic, though the cross-model validation and ablation results suggest the phenomenon is not stimulus-specific. Third, we identify Bloom filter heads based on behavioral criteria rather than a mechanistic analysis of the QK circuit weights, which would provide stronger evidence for the computational mechanism.

6 Related Work

Olsson et al. [2022] identified induction heads as a key mechanism for in-context learning, establishing the paradigm of functional head classification that we extend. Elhage et al. [2021] provided the mathematical framework for analyzing transformer circuits, including the identification of previous-token heads. Voita et al. [2019] showed that many attention heads can be pruned without performance loss, and Michel et al. [2019] demonstrated that individual heads contribute unevenly. Wang et al. [2022] traced the circuit for indirect object identification in GPT-2, identifying duplicate-token heads that attend to repeated names—the closest prior work to our Bloom filter heads, though they did not characterize these heads in terms of membership testing or connect them to Bloom filter theory. Clark et al. [2019] provided early evidence that attention heads specialize for syntactic relationships. Geva et al. [2023] analyzed factual recall circuits, complementing our focus on membership circuits.

7 Conclusion

We have identified Bloom filter heads as a new functional category of transformer attention heads, distinct from induction and previous-token heads. These heads implement approximate membership testing with the defining properties of Bloom filters: zero false negatives, capacity-dependent false positives that follow the theoretical formula, and independent hash function behavior across multiple heads. The finding generalizes across model sizes and families, and ablation confirms functional specificity.

Perhaps the most striking aspect of this finding is that gradient descent, optimizing next-token prediction loss, independently converges on a data structure that computer scientists designed by hand

over fifty years ago. Every text, Harold Bloom argued, is a response to what came before. Every token position in a transformer, it seems, begins by asking the same question: *have I seen this before?*

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A Expanded Stimulus Examples

Representative triplets from the 100-sentence stimulus set:

Exact: "The surgeon performed the operation successfully
and the surgeon spoke with the family afterward"
No-repeat: "The surgeon completed a delicate procedure lasting
seven hours before briefing anxious relatives"
Semantic: "The surgeon performed the operation successfully
and the doctor spoke with the family afterward"

B Full Per-Head Results

Complete selectivity, miss rate, and FP ratio for all 144 heads in GPT-2 small are available in the supplementary materials.